INCH-POUND

MIL-DTL-81581/8A
24 July 2006
SUPERSEDING
MIL-H-81581/8(AS)
08 December 1969

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, CABLE ASSEMBLIES, COMMUNICATION, ASSOCIATED

Reactivated after 24 July 2006 and may be used for new and existing designs and acqusition.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable assembly described herein shall consist of this specification sheet and MIL-DTL-81581.

1. SCOPE

1.1 Scope. This specification sheet covers the requirements for communication cable assemblies used with the applicable hose assemblies utilized in the Naval aviator's life support systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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FEDERAL STANDARDS

FED-STD-191	-	Textile Test Methods (Inactive for new design)
FED-STD-595	-	Colors Used in Government Procurement - Color number: 34079
FED-STD-601	-	Rubber: Sampling and Testing

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-129	-	Military Marking for Shipment and Storage

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-454	-	General Guidelines for Electronic Equipment

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWING

283AS100	-	Cable Assemblies Index, Communications

(Copies of the above drawing are available from the Naval Air Technical Data and Engineering Service Command (NATEC), P.O. Box 357031, NASNI, Bldg.90, San Diego, CA 92135-7031 or at https://webjedmics.natec.navy.mil.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from the American Society for Quality, P.O. Box 3005, Milwaukee, WI 53210-4606 or http://www.asq.org/.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO-10012-1	-	Equipment, Quality Assurance Requirements for Measuring - Part 1: Metrological Confirmation for Measuring Equipment

(Copies of this document are available from the International Organization for Standardization, American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY, USA 10036 or http://www.iso.ch/.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL) INTERNATIONAL

NCSL-Z540.1	-	Calibration Laboratories and Measuring and Test Equipment

(Copies of this document are available from the National Conference of Standards Laboratories (NCSL) International, 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404 or info@ncsli.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA-HP3	-	Electrical and Electronic Polytetrafluorethylene (PTFE) Insulated High Temperature Hook-up Wire; Types ET (250 volts), E (600 volts) and EE (1000 volts)

(Copies of this document are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209 or http://www.nema.org/.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited here in (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-DTL-81581 and as specified herein.

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3.2 Qualification. The cable furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3.1). Unless otherwise specified, each cable assembly listed on the NAVAIR Drawing 283AS100 shall be qualified separately.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3.2).

3.4 Materials and components.

3.4.1 Elastomeric material. The elastomeric material used in the fabrication of the molded or extruded portions of the cable assembly shall be of a composition which shall not bloom, shall have a minimum Shore Durometer Hardness of 55 ±5 when tested as specified in FED-STD-601, Method 3021, and shall meet the ozone resistance requirements of the general specification.

3.5 Design and construction. The communication cable assembly shall be in accordance with the applicable drawing listed on NAVAIR Drawing 283AS100 and as specified herein.

3.5.1 Communication cable. The communication cable shall consist of insulated conductors, a restraint cord (if applicable), a protective covering and termination in accordance with the applicable drawing listed on NAVAIR Drawing 283AS100.

3.5.1.1 Communication conductors. Each insulated communication conductor shall consist of 19 silver coated copper strands (No. 36 AWG) bunched with a lay of not greater than 0.75 inch. The conductor strands shall be uniformly coated in accordance with NEMA-HP3 and the conductors shall have such termination as specified on the applicable drawing. The communication conductors shall be twisted with a lay or lays, which shall be not greater than 1.75 inches.

3.5.1.2 Conductor insulation. A uniform covering of polytetrafluoroethylene (PTFE) insulating material shall be placed over each conductor in accordance with NEMA-HP3. For circuit and polarity identification, the insulation of each conductor shall have a different distinctive color throughout its length as specified on the applicable drawing.

3.5.1.3 Conductor shielding. Conductor shielding shall be as specified on the applicable drawing and shall provide a coverage of 85 to 95 percent. Shielding shall consist of tin-plated soft drawn copper (No. 40 AWG to No. 36 AWG).

3.5.1.4 Electrical fitting. When the applicable control drawing specifies the attachment of electrical fittings, such as jacks, plugs, terminal lugs, etc., the electrical fittings shall be attached to the cable in a manner which shall not impair the performance of the electrical fittings or the cable assembly. Cable assemblies having electrical fittings may be tested for insulation resistance with or without the electrical fitting attached, at the discretion of the qualifying

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activity. However, when tested with the electrical fittings attached, the specification for the electrical fitting shall govern in the event of conflict between the insulation resistance requirements specified herein and the electrical fitting requirements.

3.5.1.5 Restraint cord. The restraint cord, when specified on the applicable control drawing, shall be of a polyamide or polyester construction.

3.5.1.6 Protective covering. The protective covering shall be constructed of a concentric inner and outer layer of materials. The inner layer shall consist of a fibrous sheathing, constructed of a synthetic compound, which shall be closely wrapped, braided, or applied as a continuous tube around the shielded, twisted, insulated conductors and restraint cord (when specified). The fibrous sheathing shall in turn be tightly contained by a molded or extruded outer layer of elastomeric material which shall meet the requirements of 3.4.1. The protective covering shall be of uniform thickness and shall continuously cover the length of cable as specified on the applicable drawing.

3.6 Performance characteristics.

3.6.1 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds when tested in accordance with 4.4.2.

3.6.2 Cable tensile strength. The cable assembly shall withstand a tensile load of 35 pounds with no evidence of damage to the cable when tested as specified in 4.4.3.

3.6.3 Insulation resistance. Unless otherwise specified in the applicable control drawing, the insulation resistance of each electrical circuit shall be 100 megohms minimum when tested as specified in 4.4.4.

3.6.4 Polarity (color code) and electrical continuity. The cable assembly shall pass the polarity (color code), and in each case the electrical resistance of each circuit contained in the cable assembly shall be not greater than 0.2 ohm plus 0.03 ohm per foot length of cable when tested as specified in 4.4.5.

3.6.5 Ozone resistance. The test specimens shall show no evidence of checking, cracking, or other damage when examined under 10x magnification and tested as specified in 4.4.6.

3.6.6 Low temperature. The cable assembly shall meet the requirements of 3.6.3 and 3.6.8 when tested as specified in 4.4.7.

3.6.7 High temperature. The cable assembly shall meet the requirements of 3.6.3 and 3.6.8 when tested as specified in 4.4.8.

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3.6.8 Flexibility. The cable assembly shall have no noticeable permanent set or other damage when tested as specified in 4.4.9.

3.6.9 Flexibility endurance. The cable assembly shall be flexed without loss of conductor continuity or other damage when tested as specified in 4.4.10.

3.7 Color. Unless otherwise specified, the color of the elastomeric portions of the cable assembly shall be uniformly green, approximately matching FED-STD-595, color number 34079.

3.8 Soldering. Soldering shall follow the MIL-HDBK-454, Guideline 5.

3.9 Marking. Marking shall be in accordance with MIL-STD-129.

4. VERIFICATION

4.1 Test equipment and inspection. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of ISO-10012-1 or NCSL-Z540.1.

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room ambient temperature, relative humidity, and pressure.

4.3 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification and as specified herein.

4.3.1 Qualification inspection. The qualification inspection of the cable assembly shall consist of examinations and tests for all the requirements of this specification sheet.

4.3.1.1 Qualification samples. The qualification inspection samples shall consist of the following:

a. Four cable assemblies for which qualification is required.

b. Three test slabs 6 by 6 by 0.075 inches composed of the identical ozone-resistance composition that was used in manufacturing the outer layer of the protective covering of the cable assembly.

c. Ten 30 inch long restraint cord specimens.

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4.3.2 First article inspection. The first article inspection of the hose assembly shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.3.2.1 First article samples. The first article sample shall consist of four hose assemblies of each style contracted for.

4.3.3 Conformance inspection. The conformance inspection shall consist of the inspections specified in table I.

TABLE I. Sample size, acceptance criteria, tests and examinations of the cable assemblies.

INSPECTION	REQUIREMENT	TEST METHOD	SAMPLE SIZE 1/
Visual examination (See classification of defects)	-	4.4.1.1	Every cable assembly for critical defects. Inspection Level II for minor defects.
Flexibility	3.6.8	4.4.9	Inspection Level S-2
Cable tensile strength	3.6.2	4.4.3	Inspection Level S-2
Insulation resistance	3.6.3	4.4.4	Inspection Level S-2
Polarity and electrical continuity	3.6.4	4.4.5	Every cable assembly

1/ Sampling for acceptance inspection shall be in accordance with ASQ-Z1.4.

4.4 Inspection methods.

4.4.1 Visual examination.

4.4.1.1 Cable assembly. Every cable assembly shall be examined visually for critical defects to determine conformance to this specification and applicable drawings. The classification of defects, table II, shall be used to classify the defects found, as applicable.

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TABLE II. Classification of defects for visual examination of the cable assembly.

CRITICAL	MINOR
1. Dimensions not within specified tolerances. 2. Material imperfections - foreign matter embedded. 3. Any component missing, malformed, fractured, or otherwise damaged. 4. Any component loose, or otherwise not securely retained. 5. Incorrect assembly or improper positioning of components. 6. Any functioning part that works with difficulty. 7. Faulty workmanship or other irregularities.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent. 202. Color not as specified.

4.4.2 Restraint cord elongation and tensile strength. The restraint cord shall be tested in accordance with FED-STD-191, Method 4102. The restraint cord elongation and tensile strength shall conform to the requirements of 3.6.1.

4.4.3 Cable tensile strength. The cable assembly shall be subjected to a tensile load of 35 pounds applied at the ends of the cable acting along the longitudinal axis. The tensile load shall be maintained for one minute. The cable assembly shall conform to the requirements of 3.6.2. After completion of this test, the cable assembly shall be subjected to; meet the applicable requirements, and pass the tests specified in 4.4.4.

4.4.4 Insulation resistance. This test shall be as specified in the general specification.

4.4.5 Polarity (color code) and electrical continuity. This test shall be conducted as specified in the general specification.

4.4.6 Ozone resistance. This test shall be conducted as specified in the general specification.

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4.4.7 Low temperature. The cable assembly shall be conditioned at -65 ±5 °F for 4 hours. After the conditioning period and while still at this temperature, the cable assembly shall be subjected to the tests specified in 4.4.9. The cable assembly shall then be conditioned at room temperature and pass the tests specified in 4.4.4 and 4.4.5.

4.4.8 High temperature. The cable assembly shall be conditioned at 160 ±5 °F for 4 hours. After the conditioning period and while still at this temperature, the cable assembly shall be subjected to the tests specified in 4.4.9. The cable assembly shall be returned to room temperature and pass the tests specified in 4.4.4 and 4.4.5.

4.4.9 Flexibility. The cable assembly shall be closely coiled about a rod, one inch in diameter and then released. The test shall be repeated four times. After each release the cable assembly shall meet the requirements of 3.6.8.

4.4.10 Flexibility endurance. The cable assembly shall be coiled about a rod one inch in diameter and released. This operation shall be repeated for 500 cycles. After this test, the cable assembly shall pass the tests specified in 4.4.4 and 4.4.5. The cable assembly shall meet the requirements of 3.6.9.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cable assembly covered by this specification is intended for use by aircrew in aircraft communication.

6.2 Acquisition requirements. Acquisition requirements should be in accordance with the general specification.

6.3 Changes from previous issue The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in

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these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous issue.

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

(Project 4720-2005-019)

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